Curtin University

School of Electrical Engineering, Computing & Mathematical Sciences

Our Priority SDGs

3 GOOD HEALTH AND WELL-BEING
7 AFFORDABLE AND CLEAN ENERGY
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION

Powering T&L with Waste-to-Energy Systems:

A Dynamic Optimisation & Machine Learning Framework

An integrated, data-driven approach to advance sustainability education and innovation, combining synthetic data, machine learning, and optimisation to solve real-world energy challenges.

TEAM 1

THE GREEN ALGORITHMS

Data. Insight. Impact.

Sam Wi

Ying Li

Synthetic Data Generation Framework

Realistic, scalable, and sustainable data for robust experimentation.

+

Machine Learning Optimisation

Intelligent models that learn, predict, and adapt for better decision-making.

+

Optimisation Framework

Dynamic optimisation for resource efficiency and emission minimisation.

Curriculum Integration

Case Studies, Notebook Design, Interactive Dashboards, and more.
Empowering the next generation of sustainable problem solvers.

12 May — Phase 1: Data Collection and Exploration; Context and Industry Analysis

1 June — Phase 2: Data Engineering; Synthetic Data Generation Framework

1 July — Phase 3: Machine Learning Framework; Case Study 1

1 Aug — Phase 4: Notebook Design

1 Sep — Phase 5: Optimisation Framework; Case Study 2

1 Oct — Phase 6: Dynamic Dashboard Design

27 Oct — Final Showcase & Recognition Event

Data-driven. Sustainability-focused. Education-powered.

Building a smarter, greener future through innovation and collaboration.